Passive Silicon Mach–Zehnder Interferometers for Waveguide Group-Index Extraction

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Abstract

Five quasi-TE Mach–Zehnder interferometers (MZIs), with arm-length differences of 30, 60, 120, 240 and 420 μm, and two waveguide references are submitted for passive-silicon fabrication. The strip waveguide is 500 nm wide and 220 nm high. Each of six blocks occupies $605 \times 410\,\mu\text{m}^2$. Serialized layout centerlines determine the circuit lengths. Historical nominal EBeam models predict free spectral ranges (FSRs) near 1550 nm of 19.62, 9.73, 4.81, 2.40 and 1.37 nm. A multi-length extraction recovers a group index of 4.18972 from these simulated spectra, versus the model input 4.19088. All final upstream GitHub checks passed; PR #880 was merged. The six cells and seven measurement labels were verified in the aggregation. These are numerical and verification results; fabricated-device measurements are not yet available.

1 Objective and interferometer theory

The aim is to extract waveguide group index from wavelength-dependent interference, then compare experiment with a stated model. Varying ΔL gives a trend check and helps identify phase delay contributed by the splitters. The five designs keep the waveguide cross-section and couplers unchanged.

For propagation factor $\exp[i\beta(\lambda)L]$,

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}, \qquad n_{\text{g}} = c\frac{d\beta}{d\omega} = n_{\text{eff}} - \lambda\frac{dn_{\text{eff}}}{d\lambda}, \qquad \frac{d\beta}{d\lambda} = -\frac{2\pi n_{\text{g}}}{\lambda^2}. \tag{1}$$

Let q_1, q_2 be the complex products of splitter and combiner amplitudes, and α the power attenuation coefficient. One output has amplitude

$$t = q_1 e^{-\alpha L_1/2 + i\beta L_1} + q_2 e^{-\alpha L_2/2 + i\beta L_2}. \tag{2}$$

Thus $|t|^2$ contains interference phase $\Phi = \beta\Delta L + \phi_c$, where $\Delta L = L_2 - L_1$ and $\phi_c = \arg(q_2/q_1)$. Ideal balanced, lossless splitters give complementary outputs $T_\pm = [1 \pm \cos\Phi]/2$, up to output-port convention.

Adjacent maxima differ in phase by 2π. If the derivative is nearly constant over a fringe and coupler dispersion is negligible,

$$\text{FSR} \simeq \frac{\lambda_m^2}{n_{\text{g}}|\Delta L|}, \qquad n_{\text{g}}^{\text{app}} = \frac{\lambda_m^2}{|\Delta L|\text{FSR}}, \tag{3}$$

where λ_m is the selected peak-pair midpoint. All lengths use the same units. This is a local approximation, especially for the widest-fringe device.

For $\Delta L > 0$ and $d\Phi/d\lambda < 0$, retaining coupler phase yields

$$\frac{\lambda_m^2}{\text{FSR}} \simeq n_{\text{g}}\Delta L - \frac{\lambda_m^2}{2\pi}\frac{d\phi_c}{d\lambda}. \tag{4}$$

A fit versus ΔL estimates n_{g} from the slope while absorbing a common coupler contribution into the intercept. This assumes comparable couplers and a narrow wavelength interval; it is not a universal process-variation correction.

2 Waveguide mode and compact models

Independent field calculation

EMpy 2.2.3's full-vector finite-difference solver was used for a $0.50 \times 0.22\,\mu\mathrm{m}^2$ rectangular silicon core in homogeneous oxide [4, 3]. Propagation is along z, lateral position is x, and thickness is along y. Constant bulk indices $n_{\mathrm{Si}} = 3.4757$ and $n_{\mathrm{oxide}} = 1.446$ are explicit nominal example assumptions [4]. The domain is $2.50 \times 2.22\,\mu\mathrm{m}^2$, with zero magnetic-field outer conditions. The Maxwell eigenproblem is $\mathbf{A}(\epsilon, \lambda)\mathbf{h}_t = \beta^2 \mathbf{h}_t$; electric fields are reconstructed from its magnetic solution. No Lumerical MODE execution is claimed.

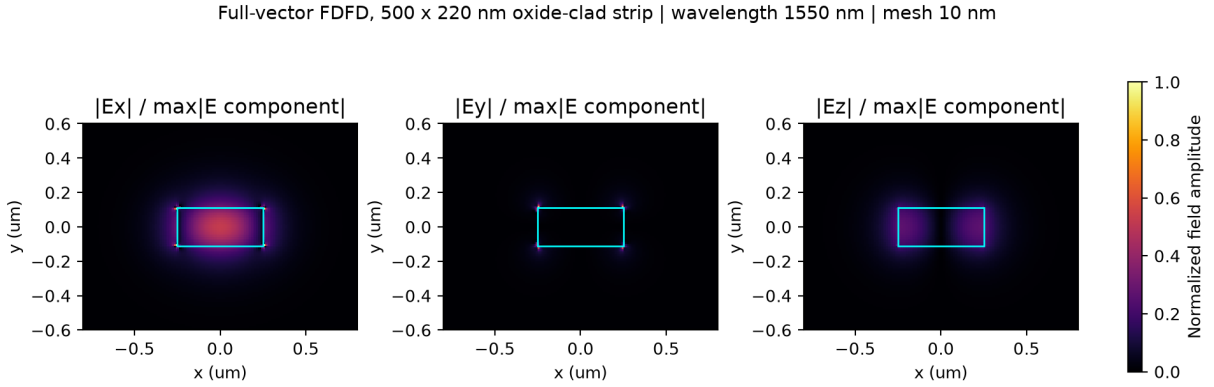


Figure 1: Fundamental quasi-TE field at 1550 nm, on a 10 nm mesh. Panels share one amplitude scale; cyan marks the silicon boundary.

Mesh (nm)	n_{eff} at 1550 nm	TE power fraction
20	2.450405	0.97754
10	2.449067	0.97970
5	2.447759	0.98089

The 10-to-5 nm index change is 0.001308 (0.0534%). A $3.50 \times 3.22\,\mu\mathrm{m}^2$ domain at 10 nm gives 2.449067242, a change of 3.9×10^{-8}. Domain truncation is small here; mesh and material approximations remain.

With $u = \lambda[\mu\mathrm{m}] - 1.55$, the five-point, 10 nm-mesh sweep gives

$$n_{\mathrm{eff}}^{\mathrm{FD}}(u) \simeq 2.4490671 - 1.0234618u - 0.12305645u^2. \tag{5}$$

Its maximum residual at the five input samples is 4.5×10^{-6}. Differentiation gives $n_{\mathrm{g}}^{\mathrm{FD}}(1550\,\mathrm{nm}) = 4.03543$. Constant bulk indices omit bulk-material dispersion, so this approximation is not substituted into the historical circuit model.

Historical PDK circuit model

The EBeam 500×220 nm TE data [2] give $n_{\mathrm{eff},0} = 2.44553$, $n_{\mathrm{g},0} = 4.19088$ at $\lambda_0 = 1550\,\mathrm{nm}$. The second-order wavelength expansion is

$$n_{\mathrm{eff}}^{\mathrm{PDK}}(u) \simeq 2.44553 - 1.12603226u - 0.03426096u^2. \tag{6}$$

Similar effective index at one wavelength does not imply a similar derivative or group index. Both models remain nominal, rather than calibrated to this fabrication run.

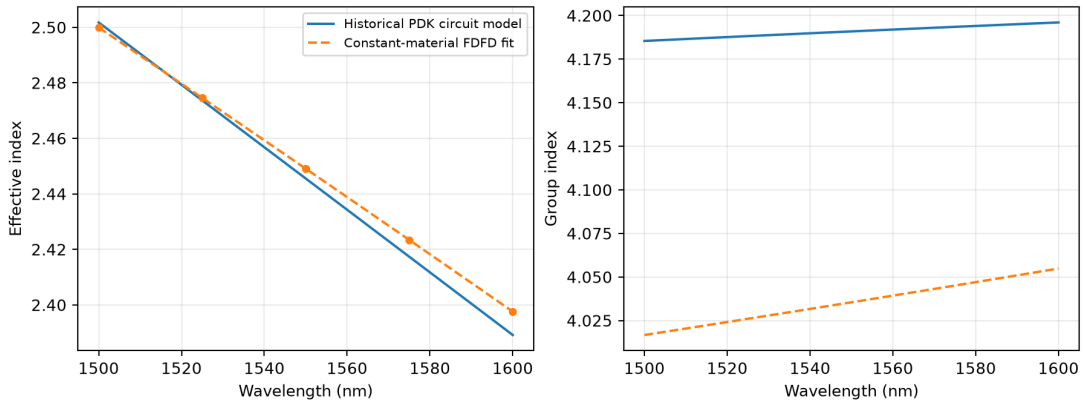


Figure 2: Effective and group indices from the two explicitly distinct models.

3 Post-layout circuit simulation and extraction

The waveguide model follows the PDK implementation:

$$\beta(\omega) = \frac{2\pi n_{\mathrm{eff},0}}{\lambda_0} + \frac{n_{\mathrm{g},0}}{c}(\omega - \omega_0) - \frac{D\lambda_0^2}{4\pi c}(\omega - \omega_0)^2, \quad D = 3.54275 \times 10^{-4}\ \mathrm{s/m^2}. \tag{7}$$

Historical Y-branch and broadband directional-coupler (BDC) complex S-parameters are interpolated in magnitude and unwrapped phase; their reflections are retained. Incoming and outgoing waves obey

$$\mathbf{a} = \mathbf{P}\mathbf{b} + \mathbf{u}, \quad \mathbf{b} = \mathbf{S}\mathbf{a}, \quad \mathbf{a} = (\mathbf{I} - \mathbf{P}\mathbf{S})^{-1}\mathbf{u}. \tag{8}$$

Here $\mathbf{P}$ connects the ports. Propagation loss is assumed to be 3 dB/cm. The 1500–1600 nm sweep uses 10 pm samples; three-point parabolic interpolation refines peak positions for the local FSR extraction.

Block	ΔL (μm)	L_1 (μm)	L_2 (μm)	FSR estimate[a] (nm)	FSR simulated (nm)	$n_{\mathrm{g}}^{\mathrm{app}}$
A	30	216.946820	246.946820	19.109	19.61927	4.04201
B	60	216.946820	276.946820	9.554	9.72914	4.11284
C	120	216.946820	336.946820	4.777	4.80957	4.15264
D	240	216.946820	456.946820	2.389	2.39957	4.17101
E	420	216.946820	636.946820	1.365	1.36866	4.17900

[a]Equation (3) at 1550 nm with $n_{\mathrm{g}} = 4.19088$. Simulated pair midpoints are 1542.413, 1549.473, 1548.124, 1549.861 and 1549.915 nm, respectively; they are not all exactly 1550 nm.

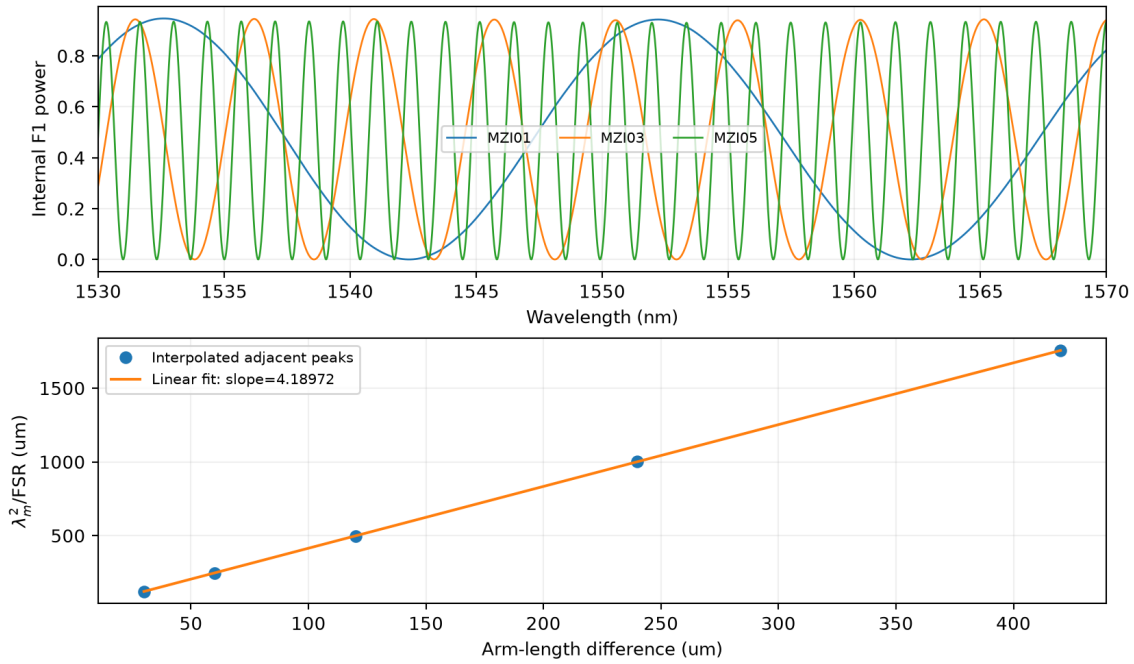


Figure 3: Three representative internal spectra and the common-intercept extraction across five designs. GC response is excluded.

Equation (4) gives a fitted slope 4.18972 and intercept $-4.49774\,\mu$m. The slope differs from the model input $n_{\mathrm{g},0} = 4.19088$ by 0.028%. This is a numerical self-consistency test, not an experimental accuracy claim. The naive single-device estimate is more biased for short ΔL, where coupler delay is a larger fraction of the total phase derivative. The FSRs are below the approximately 50 nm measurement band and far above the 1 pm minimum laser step stated in the course [1].

4 Fabrication layout and automated testing

Only silicon layer 1/0 is used for manufacturing. Heater, metal-pad and opening geometry were removed. Five MZIs occupy A–E; F contains two GC–waveguide–GC references. Each file has a distinct top name, `EBeam_linhanzhe_MZI_A` through `_F`, and its own $605 \times 410\,\mu\text{m}^2$ floorplan.

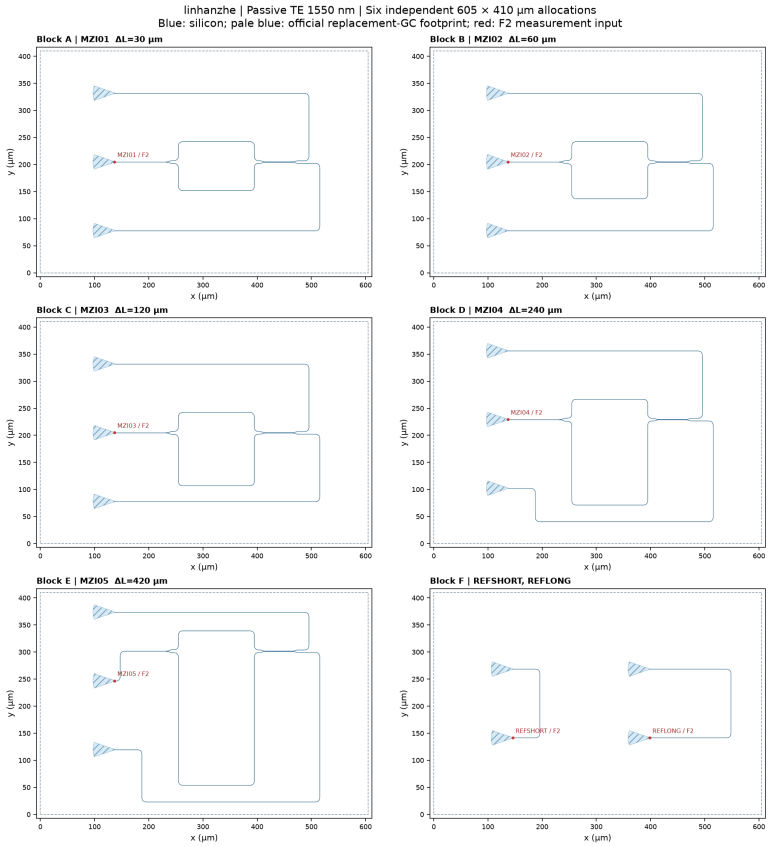


Figure 4: Six submitted blocks. Official black-box GC outlines are retained; display outlines do not represent final manufactured grating teeth.

Geometry, connections and labels

Straight waveguides are 0.5 μm wide. Euler bends have effective radius 10 μm and minimum local curvature radius approximately 7.061 μm. Physical geometry, including GC outlines, has at least 10 μm boundary clearance. The longest-imbalance MZI core was moved upward by 55 μm; no global geometric scaling was used.

Official GC, Y and BDC cells retain their original names, origins and contents. Importing PinRec paths as polygons had prevented optical-net identification. Preserving those paths and adding optical pins to generated waveguides resolves this compatibility issue. PinRec, DevRec and centerline metadata are auxiliary layers, not extra fabrication masks.

GC tips face right, with 127 μm pitch. The input aligns to F2; MZIs use F1/F2/F3 and references use F1/F2. Seven unique TEXT labels on layer 10/0 share the prefix `opt_in_TE_1550_device_linhanzhe_`, followed by `MZI01`–`MZI05`, `REFSHORT` or `REFLONG`. Each label is placed at its actual F2 input tip.

Actual lengths and reference interpretation

Actual sampled centerlines are serialized as metadata. Independent readback reconstructs the Y-to-BDC arm graph, sums the segment lengths including bends, and checks that centerline samples lie in silicon. Output leads do not enter ΔL.

Reference lengths are 220.273410 and 420.273410 μm. Their 200 μm difference gives only 0.06 dB differential propagation loss at the assumed 3 dB/cm. They support throughput and spectral-envelope comparisons. Precision propagation-loss extraction would require controlled GC variation and preferably a larger cutback-length range. No completed de-embedding or measured loss is claimed.

5 Verification, submission and measurement plan

Block	Components / nets	Submission errors	Functional markers	Manufacturing DRC markers
A	30 / 30	0	0	0
B	30 / 30	0	0	0
C	30 / 30	0	0	0
D	34 / 34	0	0	0
E	37 / 37	0	0	0
F	14 / 12	0	0	0

These local official-check results are supplemented by 12 GDS/OAS readbacks and independent 60/100 nm width/spacing checks with zero markers. Component/net counts describe recognized components and connected port nets, not that many separate circuits. Clean Python 3.11 regeneration matched the validated hierarchy and geometry.

All six native KLayout scripts ran successfully in the fork. The current upstream allowed-file rule accepts layout binaries and report PDFs, so manufacturing PR #880 contains only six OAS files. Scripts are preserved on the `klayout-source-linhanzhe` branch. PR #880 was merged on 17 October 2026 at 00:28:56 UTC+8. All seven final upstream checks succeeded, including submission, functional layout, allowed-file and final-verification checks.

The aggregation artifact from run 38932520731 was downloaded and its SHA-256 verified. Silicon, black-box and floorplan regions of all six cells exactly match the submitted files. Seven labels each occur once in the global top cell, at the following positions:

Unique label suffix	Global x (μm)	Global y (μm)
MZI01	2589.025	1874.916
MZI02	2589.025	2292.916
MZI03	2589.025	2710.916
MZI04	2589.025	3153.291
MZI05	2589.025	3588.791
REFSHORT	2597.859	3901.416
REFLONG	2850.359	3901.416

Coordinates belong to this aggregation snapshot and may change in later merges. Aggregation does not establish that final foundry GC replacement has occurred.

Measurement plan and limits

After fabrication, align F2 and record both MZI outputs and the references over the available coupling band. Find adjacent maxima using their measured midpoint wavelengths. Apply Eq. (3), inspect variation with ΔL, then fit Eq. (4) only if the shared-coupler assumptions are supported. Compare the extracted trend with the stated model and propagate wavelength/length uncertainty. Fabrication, measured spectra and calibrated comparison remain future work.

The internal simulation excludes final GC coupling, bend excess loss, external leads and process corners. It does not establish fibre-to-fibre insertion loss, yield or measured extinction ratio.

References

[1] UBCx Phot1x 3T2026, Design Proposal, Layout for Fabrication and Final Layout Submission, course materials, accessed 17 October 2026.

[2] SiEPIC, SiEPIC EBeam PDK, package 0.4.53, historical Y/BDC S-parameters and waveguide data. Source hashes are retained with this report.

[3] A. B. Fallahkhair, K. S. Li and T. E. Murphy, Vector finite difference modesolver for anisotropic dielectric waveguides, *J. Lightwave Technol.*, 26(11), 1423–1431, 2008, doi:10.1109/JLT.2008.923643.

[4] L. Bolla and contributors, EMpy, version 2.2.3; full-vector solver and mode-solver example. Material constants here are nominal assumptions.